

Ana Trišović, Ph.D.

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SUMMARY

Research Scientist at MIT FutureTech and CSAIL PI with **7+ years leading research agendas across multiple teams and institutions**. Runs **large-scale empirical studies of AI models and their real-world use**—model lifespan and supersession, benchmark–outcome relationships, and foundation-model adoption across science—via **LLM pipelines over 1M+ papers** and **methods for valid inference from classifier outputs**. Secured **\$750K+ as PI**; publishes at NeurIPS, COLM, ICLR, AIES, and *Nature SciData*.

TECHNICAL SKILLS

- **ML & AI:** model evaluation & benchmark design (predictive validity, saturation); LLM pipelines at scale (Anthropic, OpenAI, open-weight; prompting, fine-tuning, classification); classifier validation; model behavior (sycophancy)
- **Large-Scale Experimentation & Infrastructure:** million-document study design; computational reproducibility (automated re-execution of research code at scale; provenance tracking); HPC/cloud
- **Data & Methods:** Python (PyTorch, scikit-learn, statsmodels, pandas), R, SQL; inference from noisy labels (ACC/IPW, matching); causal inference (DiD, IV); measurement & reliability modeling (attenuation)

EXPERIENCE

Massachusetts Institute of Technology

Sept. 2023 – Present

Research Scientist at FutureTech, CSAIL

Cambridge, MA

- As **PI**, define the **research agenda, timelines, and resourcing** for a multi-project program on AI evaluation and AI adoption in science; secured **\$750K+** (Sloan and others); lead and mentor student and junior researchers.
- Collaborate with **Google Research** on *AI in Science: Early Insights*, and with economists, engineers, and policy stakeholders to turn findings into recommendations, including a **federal AI Request for Information**.

Harvard University

Sept. 2019 – Present

Research Associate at IQSS & HSPH

Cambridge, MA

- Led a multi-institution **large-scale study of reproducible scientific computing**: re-executed **2,000+ replication packages** and **9,000+ R scripts** (9M+ lines) and diagnosed failure causes.
- Built reproducibility automation for **Dataverse**; co-developed **SpaCE**, a causal-inference **benchmark**.

University of Chicago

Sept. 2018 – Sept. 2019

Postdoctoral Fellow at the Energy Policy Institute (EPIC)

Chicago, IL

- Engineered data models and software to estimate the **social cost of carbon** for **climate econometrics**.

CERN

July 2013 – Aug. 2018

Project Associate & Doctoral Researcher

Geneva, Switzerland

- Built **provenance and dependency-tracking** tooling in the LHCb software framework for reproducible petabyte-scale analyses; developer on **CERN Open Data**; work led to LHCb's first reproducibility policy.

RESEARCH COMMUNITY SERVICE

- **Program Committees & Reviewing:** Area Chair, AI-Native Academia workshop at **NeurIPS 2026**; Program Committee or reviewer for SciPy, NeurIPS, COLM, TaPP, ACM REP; Reproducibility Initiative Committee, SC24.
- **Editorial & Committees:** Topic Editor, **Journal of Open Source Software (JOSS)**, 2023 – present; U.S. National Committee for CODATA, National Academies; Review Board, Journal of Systems Research.
- **Invited Talks:** Swiss Federal Statistical Office, StartSe (2026); IEEE eScience, National Academies (2025).

SELECTED RESEARCH PROJECTS

Evaluating the Societal and Scientific Impacts of AI Foundation Models

2023 – Present

- Led a large-scale empirical analysis of **foundation-model adoption** in science, documenting rapid growth and a widening capability gap between frontier and research-used models ($7.7\times$ in 2013 to $26\times$ in 2024); **quantified how model scale and capability relate** to research productivity and citation impact.
- Studied compute access, funding concentration, and model adaptation capacity as drivers of AI democratization.
- Conducted global comparative analyses of AI adoption and leadership (US vs. China), comparing model scale, fine-tuning practices, and capability–impact relationships across research ecosystems.

Measuring AI in Science: Classifier Inference and Benchmark–Outcome Ceilings

2024 – Present

- Architected an **LLM-based pipeline** to classify **1M+** research papers—defining data schema, labeling framework, and **evaluation metrics**—and developed methods for **valid inference from classifier outputs** (ACC, IPW).
- Developed a **statistical framework** for when benchmark scores fail to predict real-world outcomes, decomposing benchmark–outcome correlations into **measurement reliability vs. substantive dependence**; applied to 63 open-weight LLMs and adoption across 48K papers.

EDUCATION

University of Cambridge, Newnham College

Ph.D. in Computer Science

Cambridge, United Kingdom

2014 – 2018

Union University

B.Sc. in Computer Science

Belgrade, Serbia

2010 – 2014

GRANTS & FUNDING

\$750K+ secured as PI/lead (Sloan and others); **\$7M+** as collaborator (NIH and others).

SELECT AWARDS

Google Women Techmakers Scholar | SeekCommons Fellow (STS) | 2025 Breakthrough Prize in Fundamental Physics (LHCb Collaboration) | Best Paper Award, ICML 2026 TAIG Workshop | **h-index: 19** (Google Scholar)

SELECT PUBLICATIONS

- **Ana Trišović**, Alex Fogelson, Janakan Sivaloganathan, and Neil Thompson. Science Is Falling Behind the Frontier: Foundation Model Adoption Across Half a Million Papers. *AI & Science Workshop, NeurIPS 2026*; under review, *Science Advances*.
- **Ana Trišović**. The Shrinking Lifespan of LLMs in Science. *COLM 2026*.
- **Ana Trišović** and Janakan Sivaloganathan. Open Science, Closed Models: How Funding Shapes AI in Science. *AIES 2026*.
- Kristina Sakayeva and **Ana Trišović**. Reproducing ELEPHANT: Judge-Dependence and Model Deprecation in an LLM Sycophancy Benchmark. *Trust-AI-Eval Workshop, NeurIPS 2026*.
- Hans Gundlach, Alex Fogelson, Jayson Lynch, **Ana Trišović**, Jonathan S. Rosenfeld, Anmol Sandhu, and Neil Thompson. On the Origin of Algorithmic Progress in AI. *NeurIPS 2026* (Best Paper, ICML 2026 TAIG).
- Mihai Codreanu, Alex Imas, Juan Mateos-Garcia, Joseph Emmens, Evalyne Muiruri, Arthur Turrell, Julian Jacobs, Atoosa Kasirzadeh, **Ana Trišović**, Yiyuan Chen, Tanya Rodchenko, Catherine Pollard, Scott Strand, Daniel Rock, Zanna Iscenko, Fabien Curto Millet, Neil Thompson, and James Manyika. AI in Science: Early Insights. *Technical report, Google*, 2026. <https://ai.google/static/documents/AI-in-Science.pdf>
- Mauricio Tec, **Ana Trišović**, Michelle Audirac, Sophie Woodward, Jie Kate Hu, Naeem Khoshnevis, and Francesca Dominici. SpaCE: The Spatial Confounding Environment. *ICLR 2024*.
- **Ana Trišović**, Matthew K. Lau, Thomas Pasquier, and Mercè Crosas. A Large-scale Study on Research Code Quality and Execution. *Nature Scientific Data*, 2022.